Chapter 10

Medical Literature Evaluation and Biostatistics

Christopher S. Wisniewski, Emily P. Jones, Erin R. Weeda, Nicole A. Pilch and Mary Frances Picone
Medical University of South Carolina, Charleston, SC, United States

Learning Objectives:

Objective 10.1 Overview medical literature evaluation and common features of reputed journals.
Objective 10.2 Explain how to interpret and evaluate a typical journal article, including the common biostatistics methods.
Objective 10.2 Utilize clinical expertise to identify the overall validity of a given clinical trial.

OBJECTIVE 10.1. OVERVIEW MEDICAL LITERATURE EVALUATION AND COMMON FEATURES OF REPUTED JOURNALS

Medical literature evaluation (MLE), defined as reading, comprehending, summarizing, and critiquing literature, is a required skill for pharmacists.[1–4] Specific components of this skill set include critically analyzing literature, summarizing basic study design and biostatistics methodology, and differentiating between statistical and clinical significance.[1,2]

These skills are necessary regardless of the setting, as evidenced by the fact that pharmacy organizations recognize MLE competence as an entry-level ability for pharmacists.[5–7] MLE skill is useful in education, practice, and research; anything from answering a simple query using a journal article to preparing a systematic review to clarify a complicated clinical controversy requires MLE skills. In this era of information overload, it is necessary for pharmacists to develop their MLE skills, including selecting, assessing, and interpreting relevant literature. Each time a pharmacy student prepares assignments involving literature review, MLE skills will improve.

Journals

Prior to appraising an article itself, one must first consider the journal. There are many aspects to assess when evaluating a journal's reputation (Table 10.1). The most important is whether the journal has a peer-review process, which is an assessment by outside experts who critique the manuscript prior to publication.[8,9] Advantages associated with peer review include consideration of the article's appropriateness for the journal and the opportunity for the author to receive feedback from other experts in the field in order to improve their work. Peer review does introduce lag time, a delay in publication, and the quality of the review and reviewer is directly proportional, so peer review is not a guaranteed process for quality assurance. Peer review is considered a gold-standard component of the medical literature publication process and is a major qualifying factor for journal reputability, but peer review does not mean that a paper is beyond reproach; regardless of whether peer review has been conducted, careful consideration should be taken when reading an article.

A relatively new concept, seemingly an alternative to peer review, is open access publishing.[10,11] This is a publisher policy which makes articles available free to readers, allowing for duplication, distribution, and use if the source is credited. True open access does not use the formalized peer review because of the theory that poor articles will be identified by the medical community following publication and subsequently ignored.

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TABLE 10.1 Considerations in the Evaluation of Journal Reputation

• Purpose of journal (i.e., why is it published)	• Organization association
• Types of medical literature published (i.e., original research or review articles)	• Source (i.e., country of origin) • Editorial board and policies
• Financial support	• Publisher

Identifiers and Metrics

Identifiers are utilized at the journal, author, and object level to aid in attribution and to combat ambiguity. Bibliometrics also applied to the journal, author, and article levels are quantitative measures of scholarly impact that help researchers and organizations make decisions regarding scholarly output. Because no one metric is comprehensive and each has its own limitations and strengths, it is highly recommended that multiple bibliometric performance indicators be applied as a group.[12]

The primary journal identifier is the International Standard Serial Number (ISSN). Used to find serialized publications such as journals, databases, and websites, ISSN is similar to International Standard Book Number, a unique identifier assigned to books.

Metrics have been developed that assess the influence of journals. These metrics may affect researchers' preferences about where to submit manuscripts for publication. There are a variety of journal metrics and each has its own algorithm with its own strengths and limitations (Table 10.2).

The most well-established journal metric, journal impact factor (JIF), reflects the frequency of a journal's publications cited in the literature.[13] This is determined by counting the number of times articles published in the preceding 2 years were cited in the current year and dividing that by the total number of articles published in the journal during those two prior years.[13] For example, the number of times that articles published in a specific journal in 2015 and 2016 were cited in 2017 would be divided by the total number of articles that journal published in 2015 and 2016. SCImago journal rank considers articles published in the previous 3 years, whereas 5-year impact factor considers 5 years. These metrics do not consider the immediate citations received in the same year of publication. Immediacy index, calculated by dividing the number of citations over the number of articles published in the current year, indicates the immediate impact of the journal. Conversely, some journals routinely publish articles that are highly cited for a few years and then ignored. Cited half-life measures the number of years, going back from the current year, that account for half the total citations received by the cited journal in the current year; a long cited half-life means the journal has a good track record of publishing useful articles in the past.

There are several other metrics used to evaluate the dissemination and utilization of research published in a specific journal. Table 10.2 defines these tools and provides some additional information on their value.

The articles themselves have identification tools and are measured via bibliometrics. A digital object identifier (DOI) is a unique alphanumeric code assigned to publications and is regulated by the International DOI Foundation. If known, DOIs allow for easier retrieval of articles and other born-digital publications and act as an online address for the lifetime of the article.[21]

Crossref is an official DOI registration agency that interlinks online content.[22] Many journals publish "online-first" versions of articles before completing the editorial process to provide early access to readers. Crossmark is an option usually found with online articles, by clicking, that will suggest if the version is current or not.[23]

Article-level metrics attempt to estimate the impact of an individual research article. Originally, article-level metrics derived from citations counts because if an article is cited many times, the implication is that the article was useful.[24] Advances in technology have led researchers to investigate using alternative metrics, or altmetrics, for a more comprehensive view of research impact beyond the bounds of traditional, scholarly publications. Altmetrics are defined as "a set of methods based on the social web to measure, track, and analyze the scholarly output."[25] Altmetrics goes beyond traditional citation counts for article-level metrics and uses social web activity to estimate the impact through

TABLE 10.2 Journal Bibliometrics[13–19]

Journal Metric	Measures	Strengths	Limitations	Found In
Journal impact factor (JIF)	Journal prestige within a particular discipline	• An established metric; been in use the longest • Depending on the subject, more journals will be indexed in Web of Science than Scopus allowing for more JIF scores • Easy to calculate • Related metrics include ◦ 5-year Impact Factor ◦ Cited-Half Life ◦ Immediacy Index	• Need a subscription to Journal Citation Reports • Citations are not weighted • Scores not universal among all disciplines • Includes all self-citations, which raises concerns of skewing impact • Takes a minimum of 6 years for a newly published journal to receive a JIF • Report on very few open access journals[a]	Journal citation reports (Web of Science)
Eigenfactor scores	Journal importance	• Freely available • Weighted citations • Algorithm accounts for citation differences of disciplines	• Excludes all self-citations which can have a negative impact on fields with limited researchers where self-citation is necessary • 5-year citation analysis period can be too long for fast-moving fields[13]	Eigenfactor.org
SCImago journal rank (SJR)	Journal influence	• Freely available • Weighted citations • SNIP tool • Report on open access journals • Includes a limited number of self-citations to account for impact measurement issues surrounding reporting them on an all or nothing scale	• Updated twice per year, whereas JIF is updated yearly • Only peer-reviewed citations are counted; can be a limitation for some authors	Scopus SCImagojr.com
CiteScore	Journal impact	• Freely available • Updated monthly • Comprehensive • Transparency • New publications receive CiteScore metrics the year after being indexed by Scopus	• New metric-less research on the metric • Potential conflict of interest	Scopus

SNIP, source normalized impact per paper.
[a]*In June 2017, Clarivate Analytics announced a partnership with Impact Story to include 18 million open access articles in Web of Science.*[20]

clicks or views, downloads, saves, and online mentions. Like other metrics, altmetrics are not meant to be used as the sole indicator of the impact. However, they should be used in conjunction with other metrics to have a more comprehensive view of the impact.

Research Reporting Guidelines

Over time, individuals have collaborated to develop a standardized research reporting structure for writing manuscripts related to various study designs. The first of these, Consolidated Standards of Reporting Trials (CONSORT), was created in 1996 to address the lack of adequate reporting of randomized controlled trials.[26] These are listed in Table 10.3, along with a brief description.

TABLE 10.3 Manuscript Reporting Guidelines for Study Designs Published in the Medical Literature[27]

Study Design	Description	Reporting Guideline
Case report	Description of clinical course of a single patient	CARE
Observational	Investigate naturally occurring variables and outcomes to identify associations	STROBE
Randomized controlled trial	Establish cause and effect through direct interventional comparison with control for potential confounders	CONSORT
Systematic review	Study and analysis of primary literature identified via a structured process	PRISMA
Clinical practice guidelines	Review and evaluation of evidence summarized to inform clinicians of best practices for patient care	AGREE
Economic evaluation	Outcomes research considering quality and cost of therapy, typically via pharmacoeconomic research	CHEERS

OBJECTIVE 10.2. EXPLAIN HOW TO INTERPRET AND EVALUATE A TYPICAL JOURNAL ARTICLE, INCLUDING THE COMMON BIOSTATISTICS METHODS

Title of the Article

After considering the journal, evaluating the paper begins. The title should be reflective of the work, unbiased, and specific yet concise.[9] Titles should not be declarative sentences, which can overemphasize results, or questions. Including the study design as a subtitle can be helpful. When encountering acronyms, one needs to be wary of duplicate use, rule distortion, and coercion. For example, an article titled "**A**zi**l**imide post**i**nfarct sur**v**ival **e**valuation" used the acronym "ALIVE,"[28] which is a common acronym in the medical literature, distorts the rules by using letters that do not start words in the title, and could be considered coercive because the acronym implies that participants would stay alive if they enrolled in the study. Acronyms do have value, as they can simplify reference to and recall of articles for clinicians, but authors should consider the development of study titles to ensure appropriate use of acronyms.

Authors

Authorship, which is defined as a substantial contribution to a publication, should be considered from two perspectives.[8] As a reader, it is important to evaluate author credentials specifically; profession, degree, training, and practice site.[9] For example, a community pharmacist might not be qualified to write a paper on hospital reimbursement procedures. Reviewing previous publications by the author is another mechanism to determine if an author is qualified to write on a subject. One could use author-level metrics, which measure individual researcher scholarly productivity and impact, when conducting evaluations. The most common author metrics are h-index, g-index, and the i10-index (Table 10.4).[17,29]

From a researcher perspective, when initiating a project that is intended for publication, initial conversation among collaborators should involve the place of authorship. Roles that are generally considered appropriate for authorship are

TABLE 10.4 Author-Level Bibliometrics[17,29]

Author Metric	Strengths	Limitations	Found in
h-index	• Most commonly used	• H-index scores are often different in each database because they are highly sensitive to database coverage • All citations are equal	Scopus Google Scholar Web of Science
g-index	• Accounts for performance or quality	• Fairly new • Not as widely accepted	Publish or Perish tool
I10-index	• Weights highly cited articles	• Only in Google Scholar • Limited to journal offerings within Google Scholar	Google Scholar

study conception and design, statistical analysis, and writing.[8] The first author is the person who completes most of the writing. The last author is the conceiver of the project and takes on a broad oversight role. Authorship is not indicated for people responsible for acquisition of funding or research group supervision.

Unethical practices are found to affect the quality of scientific communications. When authors plagiarize, fabricate, or falsify information, this distorts the integrity of research and introduces a conflict of interest. Conflicts of interest may occur due to funds received or an intention to increase a professional profile. The demand to publish (i.e., publish or perish) also increases research fraud. A common practice is self-plagiarism, which occurs when authors publish the same content in more than one journal without permission from the first journal.[30]

Guidelines found in journals explain the best practices to authors, reviewers, and editors, and journals instruct key points for researchers and authors. The authors should sign a contract with the publisher to avoid any unethical practices in publishing. Standardization of articles is mandated by the research reporting guidelines listed in Table 10.3.[27]

Author identifiers allow researchers to distinguish themselves from others and increase proper attribution of publications; these are especially helpful against combating ambiguity issues such as authors publishing under many name variations (e.g., Smith, J., Smith, J.C., or Smith, John), common names, name changes, and cultural differences in naming. Databases such as Scopus and Web of Science have integrated author identifiers into their records using Scopus Author Identifier and ResearcherID, respectively.[31,32] Open Researcher and Contributor Identification (ORCID) is a nonprofit, globally recognized digital identifier.[33] ORCIDs are free to obtain, and authors can add their Scopus and Web of Science identifier to their ORCID profile and vice versa.

Abstract

The next section of the evaluation, the abstract, is a concise summary of the main points of an article that usually addresses the background, methods, results, and conclusions of a paper. The primary function of the abstract is that it allows a reader to identify if they should read the article; it is also utilized to determine relevance when conducting a literature search.

From a reader perspective, the abstract should be thorough, concise, unbiased, and consistent with the manuscript.[9] Authors are often unable to report all study outcomes in the abstract because of limitations on word count. Furthermore, the authors occasionally publish abstracts that differ from the full paper, particularly in conclusion. For these reasons, no therapeutic decision should ever be made solely based on the information in the abstract, assuming that abstract is associated with a full article.

Introduction

The introduction of a paper provides background on the topic and states the article's purpose.[8] Starting with a brief review of the issue to educate the reader, the introduction then includes a review of the literature and a rationale for how the study will resolve unanswered questions. The introduction should end with a specific objective explaining the goals of the study. As a reader, the introduction is valuable to learn about the area of the study, but one should read this section carefully, being on the lookout for selective references and biased wording.

Methods

The aim, objective, or goal (all of these terms can be used interchangeably) the authors present at the end of the introduction should directly inform the study design that they select. As the consumer of the written research publication, careful evaluation of the study design is needed. The presented aim should lead to a study design that makes sense.

Study Design

Table 10.5 lists common study designs, their use, and advantages and disadvantages organized by the hierarchy (Fig. 10.1) as it relates to the ability to identify cause and effect. This section will describe elements of controlled trials that must be evaluated but could be extrapolated to other study designs.

Study Design Objectives

When conducting controlled trials, there are three ways in which a comparison is made, usually referred to as the objective of the study design. These include superiority, equivalence, and noninferiority.

TABLE 10.5 Study Design Uses, Advantages, and Disadvantages

Study Design	Used to:	Advantages	Disadvantages
Case report	Describe clinical course of a patient	• Quick and inexpensive • Identify treatments for rare disorders • Medication toxicities or adverse effects • Initial step for generating hypotheses	• Small sample size • Results could be arbitrary • Author opinion • Confounders
Cross-sectional	Identify prevalence	• Quick and inexpensive • Multiple outcomes • One-time data collection • No loss to follow-up • Focus on precise populations • Lack ethical difficulties • Hypothesis-generating • Can repeat to show a trend • Oversample subgroups	• Only one point in time • Cannot differentiate cause and effect from association • Transient effect potential • Difficulty determining the sequence of events • Potential for bias (e.g., selection, volunteer, recall) • Low response rate potential • Inaccurate for rare diseases
Case-control	Identify risk factors for diseases	• Efficient for rare outcomes • Quick and inexpensive • Data already exist • No loss to follow-up • Lack ethical difficulties • Consider a large number of variables • Hypothesis-generating	• Retrospective design (e.g., selection of controls, determination of exposure) • Cannot determine cause and effect • Cannot calculate incidence or prevalence • Only one outcome • Confounding variables • Difficulty determining the sequence of events • The potential for bias (e.g., sampling, recall)
Cohort	Identifies incidence or natural history of a condition	• Prospective-demonstrate exposure preceded outcome • Multiple outcome variables • Lack ethical difficulties • Outcomes from rare exposures	• Prospective-expensive • Prospective-a loss to follow-up • Multiple variables cause confounding factors • Prospective-the potential for selection bias • Retrospective-the potential for recall bias • Both-potential for surveillance bias • Not optimal for rare outcomes
Controlled trials	Evaluates efficacy	• Demonstrates exposure preceded outcome • Randomization eliminates potential confounders • Typically one exposure	• Expensive and time-consuming • Potential for loss to follow-up • Ethical constraints • Multiple outcome variables • Inefficient for rare exposures and outcomes
Systematic reviews/ metaanalysis	Identify primary literature via systematic search and evaluate or combine to answer clinical controversies	• Quick and inexpensive • Increases comprehension of the available evidence • Useful for contradictory studies • Data already exist • Lack ethical difficulties • Metaanalysis-increases power	• Compilation of data is only as valuable as data identified and published • Data not generated for reanalysis purposes • The potential for "gray" (i.e., unpublished) literature to introduce publication bias • Language biasconsidering literature for inclusion based on authors' ability to read/write in that language

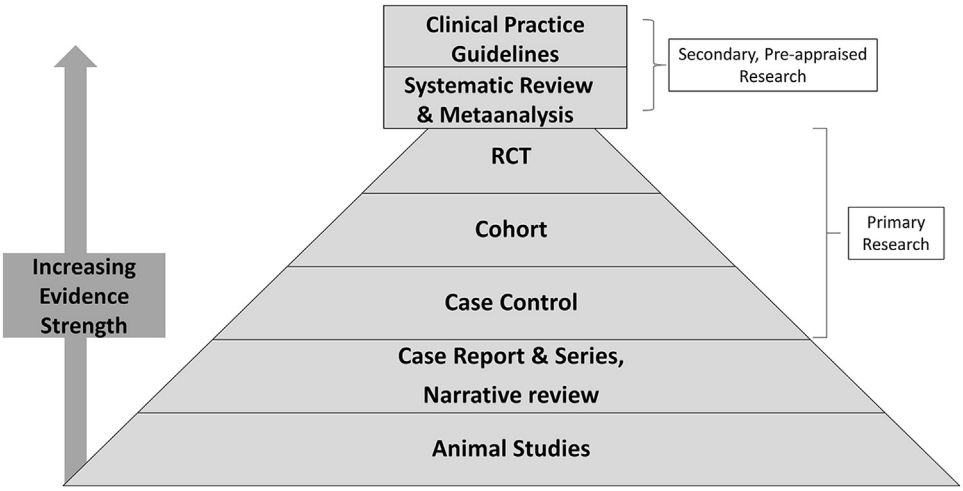


FIGURE 10.1 Hierarchy of study designs in generating evidence.

If the objective of the study is superiority, then the study is trying to prove that something is better than something else. This objective is utilized to demonstrate that a novel intervention is better than either a placebo or active control. Equivalence studies are those studies in which groups are compared with the aim of demonstrating that the effect of one intervention is equal to another, meaning that the amount of response is clinically unimportant. Frequently, equivalence trials in the field of pharmacy are coined bioequivalence and are the design used in trials to compare brand and generic medications. Noninferiority studies look to see if an intervention can produce similar results without being worse than the standard therapy.[34] For example, a medication that is used to prevent rejection in kidney transplant patients and has been used for several years could serve as a control in a study. A novel agent is developed that is similar to the parent compound but causes less gastrointestinal side effects; however, the new medication is thought to be as effective as the standard medication. The noninferiority objective would be appropriate to prove that the new drug is no worse than the standard from an efficacy perspective.

Setting

When considering the study setting, there are three types: inpatient, outpatient, and artificial. The inpatient setting includes hospitalized patients and provides the advantage that patients are located in a predictable setting. This can lead to improved compliance (e.g., patients taking medications as prescribed), less protocol variation, and increased opportunity for evaluation. This setting is ideal to conduct studies that are short and only require observations while the patient is inpatient. The inpatient setting can be difficult if recruiting patients who are admitted for nonelective reasons. For example, studying a new antimicrobial agent for sepsis would seem to be feasible; however, obtaining consent in times of duress is challenging. Additional disadvantages include the inability to predict discharge and coordination with the medical team.

The outpatient setting is used when patients experience interventions in the context of their normal daily activities. This setting has several advantages, mainly that it is more natural and less expensive, but the direct oversight is lost, leading to issues with compliance and patient-driven data collection.

Artificial settings allow for research in a standardized way and are frequently used for convenience and early interventions. These types of settings can achieve the research objective quickly as they are not impacted as much by patients; however, they often do not include what will happen if an intervention is used on patients. This type of environment is often limited to early studies and is not commonly seen in the pharmacy literature.

Readers should be careful when applying the findings of studies across settings. For example, using results from an inpatient study might cause problems when the clinician tries to relate them to nonhospitalized patients due to problems with observation and compliance. It is important to ask how the results of a study in a given setting might be impacted by using the intervention in a different setting before making patient-care decisions.

Patient/Subject Selection

When enrolling subjects, two groups must be established: the population, which includes all objects of a particular kind, and a sample, a subset that will be used to represent the population. During sample selection, researchers want to make sure

that they can make generalizations from results and extrapolate them to the population. This requires investigators to get a representative sample and use statistics to measure the results and apply them to the population.

After sample identification and potential participant selection, screening can take place. Investigators use inclusion and exclusion criteria established during protocol development. Inclusion criteria define the characteristics needed for a potential participant to be enrolled. For example, in a study evaluating hypertension, the individual must have higher than a specific blood pressure. Exclusion criteria are characteristics that prevent enrollment in a study; these can be for clinical, safety, or likelihood-to-withdrawal reasons.

Once a subject meets the enrollment criteria, consent must be obtained. When screening patients, it is important for the participant to have a working knowledge of the goals of the study and their own risks and benefits. Rules that direct how one can approach patients are governed by local institutional review boards (IRBs), which are dictated based on international and national requirements related to past human abuses (e.g., Declaration of Helsinki). The IRB is responsible for reviewing all human subject research and ensuring that the protocol and related study activities, including informed consent, are conducted in an ethical manner.[35]

As a reader, it is important to consider whether the sampling techniques used would lead to a sample representative of the population. The inclusion and exclusion criteria should be considered to ensure that enrolled patients will yield valid results and bear similarity to patients that are under one's care. All studies should have IRB approval and require informed consent; if not mentioned, the study should be disregarded.

Patient Handling

Following enrollment, studies are organized to compare participants, usually via intervention and control groups. Interventions in the pharmacy literature are pharmacologic in nature. Goals of therapy, the dosage, dosage form, and regimen, and any concurrent medications allowed should be considered when evaluating an intervention. For example, a suspension must be shaken prior to administration, which may lead to inconsistency in exposure versus a liquid. To overcome this, an investigator could provide the recipient with prefilled syringes to ensure that the correct amount of drug is in each dose. If that suspension interacts with grapefruit juice, did the investigators tell their participants to avoid grapefruit juice? The article should detail these nuances.

Study subjects receiving the intervention are compared with a control group, thus the name controlled trials. Several types of controls are used in the pharmacy literature. Historical controls are previously collected data from patients with similar baseline demographics. A placebo control is an inert entity that is otherwise identical in appearance, taste, and smell to the intervention. Placebo-controlled trials are the minimum requirement for approval by US Food and Drug Administration but should only be used for conditions in which there is not an acceptable therapeutic alternative. An active control is often used when comparing a newer intervention to the standard therapy; both arms receive an active intervention. Active controls are used when forgoing a treatment is unethical or when noninferiority is the objective. Some studies will use no treatment as a control, allowing researchers to show the true efficacy of the intervention.

From a reader perspective, make sure the intervention makes sense, therapeutically and for specific patients one sees. Although using no treatment as a control is the best choice for establishing efficacy, the most common controls are either placebo or active. Placebo controls reduce the risk of bias and are useful to maintain blinding, but may decrease participation or lead to patients overestimating efficacy due to the belief that they received therapy, commonly referred to as the placebo effect. As a clinician, active controls are preferable, but because most of these studies use a noninferiority objective, the true effect of the intervention is difficult to ascertain.

Once patients are selected for inclusion in the study, they will be assigned to the intervention or control groups. There are three strategies for assigning patients: randomization, blinding, and allocation concealment.

Randomization is the principle that elevates controlled trials above observational study designs because randomization gives participants an equal chance of being placed in either group. This minimizes bias by prioritizing similar groups, assuring the only difference between groups is study treatment (i.e., intervention or control).

Blinding prevents subjects or investigators from knowing the treatment status of participants. Blinding reduces bias, limits altered expectations, and increases compliance because patients know there is a chance they are receiving the control. Blinding is especially important for subjective outcomes (e.g., pain), as patients may feel they are benefitting even though they are receiving control. Blinding is less important for objective outcomes (e.g., mortality) where the definition of the outcome is not up to the patient. Double-blind, the gold standard, refers to studies where both the investigator and participant are blinded; other types include open-label (i.e., no blinding), which is acceptable for studies using objective outcomes, single-blind (i.e., one group blinded), and triple-blind (i.e., investigators, participants, and assessors blinded).

Allocation concealment, the third assignment technique, is a procedure where the allocation sequence is not revealed to those assigning participants and thus prevents study personnel from influencing group assignment. Once the randomization sequence has been executed, the enrollee should be assigned to whichever group is next; otherwise, individuals could identify the assignment and push the next patient into the intervention group for the patient to receive therapy instead of control (e.g., placebo).[36] This would cause unequal groups even though technically the randomization sequence was executed correctly.

In randomized trials, patients are followed via either a parallel or crossover methodology. Parallel-group studies enroll patients into treatment or control groups, and they receive only that option. Groups are followed forward in time and then compared against each other. Crossover studies allow participants to serve as their own control by assigning patients to one intervention and then switching them to another intervention later in the trial.

The best-designed studies use all three assignment techniques. While reading an article, check that the assignment techniques investigators use make sense and are appropriate. For example, open-label studies with subjective outcomes likely have decreased value. Parallel study designs are more common in the pharmacy literature than crossover studies. For crossover studies, check that investigators have incorporated a washout period, which is a time that the participant receives no therapy and is considered long enough for the drug to clear from a person's system. This decreases the possibility that the previous therapy would have a continued effect after the new therapy was started.

Data Handling

To understand data handling, three terms must first be defined. Efficacy describes how well an intervention works in an ideal environment, whereas effectiveness describes how well an intervention works in a real-world setting.[37] Finally, attrition occurs when participants are lost to follow-up or withdraw from a study.[38] Attrition occurs if a participant is noncompliant or experiences an adverse event or therapeutic failure while receiving study treatment.

There are three main analyses to handle data: intention-to-treat (ITT), per-protocol (PP), and modified intention-to-treat (mITT).[39–41] In an ITT analysis, all subjects are analyzed even if attrition occurs. In contrast, in PP analysis, patients experiencing attrition are excluded from analysis. An mITT analysis combines principles used in ITT and PP analysis; though the definition differs across studies, most commonly mITT includes patients receiving at least one dose of the study medication.

ITT analysis is advantageous because it preserves randomization[39–41]; thus, the observed treatment effect is more likely a result of the intervention, not an underlying difference in study group characteristics. Another advantage of ITT is that it more closely resembles clinical practice because some patients comply with medication regimens and some do not. Therefore, compared with PP analysis, ITT analysis allows for improved measurement of the effectiveness of a treatment. A disadvantage of ITT analysis is that the treatment effect could be attenuated by noncompliance; an intervention will have a lesser impact in the included noncompliant patients, meaning the effect may be smaller than what would be observed if only compliant patients were analyzed. The key advantage of PP analysis is that it measures efficacy.

Study Outcomes

The outcomes of a study are classified in several ways. First, there is the distinction between primary (i.e., those that measure the most important question in a study) and secondary.[42] Decisions about study methods will be dictated by the expected effect of the intervention on the primary endpoint. In contrast, secondary endpoints are often included to support the primary endpoint.

Endpoints are often considered as surrogate or final health outcomes.[43] Final health outcomes, sometimes called clinically meaningful outcomes, directly measure survival or patient functionality. These outcomes include clinical events (e.g., myocardial infarction [MI]), health-related quality-of-life, and mortality. In contrast, surrogate outcomes are physical signs or laboratory values that serve as an indirect measure of clinically meaningful outcomes. For instance, increased blood glucose is associated with microvascular complications in patients with diabetes.[44] In a study of a medication for diabetes, an endpoint measuring microvascular complications would be considered a clinical outcome, whereas an endpoint measuring the ability of the medication to lower blood glucose would be a surrogate outcome.[43] Surrogate outcomes can be collected faster than clinical outcomes and are used to save time and money.[45] For instance, changes in blood glucose can be measured within 3 months, whereas measuring the microvascular complication of reduced kidney function may require an investigator to follow patients for years. A key disadvantage of a surrogate outcome is that it may not correlate with the final health outcome of interest. There are numerous examples of medications that positively impacted surrogate outcomes but were later shown to have detrimental effects on final health outcomes.[45] If a surrogate outcome is used, the reader should assess if there is an established relationship between the surrogate and clinical outcome.

Studies may utilize a composite endpoint, which combines multiple endpoints into one measure.[42] For instance, in a cardiology trial, the primary endpoint could be a combination of MI, stroke, or cardiovascular death. As compared with designating only one of these events as the primary endpoint, the number of patients experiencing this composite endpoint will be larger. This requires fewer patients and shorter follow-up; however, if a large discrepancy in the occurrence of individual events occurs, composite endpoints may be compromised, especially if the most clinically important event occurs less frequently. Assuming cardiovascular death is most impactful, if the majority of patients in the study are counted as experiencing the composite endpoint because they had a MI, conclusions about the impact of the medication on cardiac death can be difficult to make.

Follow-Up Procedures and Sample Size

To assess the impact of an intervention on outcomes, readers must consider both follow-up and sample size.[46–49] Researchers should choose a length and frequency of follow-up that allows outcomes to be accurately captured.[46] For example, if an antidepressant is not fully effective until 1 month after use, a study that only follows patients for 1 week would not accurately capture the impact of the medication. Readers should also bear in mind that the longer the follow-up, the greater the risk of attrition. Finally, readers should assess compliance with the treatment during follow-up.[47] If compliance was low, the authors should explain (e.g., tolerability of the medication) in order to assist readers in assessing the benefit and risk of the intervention.

To detect differences between treatment and control groups, an adequate sample size is needed.[49] The smaller the difference between treatment and control, the larger the sample size needed to detect this difference. Similarly, when events occur infrequently, more patients will have to be studied to capture these events. Sample size can be limited by cost constraints (as the cost can increase with study size) and other logistical challenges of obtaining data on a large number of participants.

Variables

Variables (i.e., the characteristics measured in a study) are classified as either independent or dependent.[50] Independent variables are those altered by an investigator because of their expected impact on the dependent variable. If an investigator is studying the impact of a low-fat diet on heart disease, diet is the independent variable because it can be manipulated to measure the impact on the dependent variable of heart disease.

The relationship between variables can be distorted by confounders.[51] Confounding occurs when a factor related to the variables of interest meets the following criteria: (1) it impacts the outcome and (2) it is unevenly distributed among treatment and control groups. A confounding variable can lead to an association between the treatment and outcome when one does not exist. For instance, participant exercise performance could influence the results of the heart disease study if subjects in the low-fat diet workout more often than those in the control group. Readers should assess if authors attempted to minimize the impact of confounding either in the design (e.g., assignment techniques) or analysis (e.g., statistical adjustment through techniques such as regression) of the study.[52]

Data

Variables are classified as either nominal, ordinal, continuous, or time-to-event.[50] Nominal data are divided into named, unranked categories that are mutually exclusive. Examples of nominal data include sex, race, or presence of a condition (e.g., diabetes or no diabetes). Ordinal data also fall into categories but differ from nominal data in that these categories are ranked. Stages of disease (e.g., stages I–IV cancer) are often ordinal data. An important characteristic of ordinal data is that the increments between categories are unequal. In the example of cancer stage, stage III has a poorer prognosis than stage II, which has a poorer prognosis than stage I; however, the difference in prognosis between stage III and stage II versus stage II and stage I is not equal. Continuous data do not fall into categories; rather, these data occupy a range of values. Examples of continuous data include blood pressure, blood glucose, weight, and temperature. In studies where outcomes may occur at any point during follow-up, researchers may wish to explore if the intervention leads to differences in the time to the development of the outcome; therefore, time-to-event data are analyzed.[53] Time to the progression of a disease, resolution of symptoms, or death are all examples of such data.

Data are also organized by the manner in which they were generated for comparison purposes. Independent data are derived from separate groups under different conditions, typically via a parallel study design. Paired data are derived from the same person or group under two different conditions. Crossover studies or comparisons of baseline data with the data generated after an intervention are good examples of paired data.

TABLE 10.6 Descriptive Statistics Definitions and Application Based on Data Type

Descriptive Measure	Definition	Type of Data Appropriate for Use
Mean	Average of all values	Normally distributed continuous data
Median	Middle data point in set of ranked data	Ordinal, continuous
Mode	Most frequently occurring value	Nominal, ordinal, continuous
Standard deviation	Average amount each data point is from mean	Normally distributed continuous data
Range	Distance between highest and lowest value	Ordinal, continuous
Interquartile range	Distance between 25th and 75th percentile values	Ordinal, continuous

Descriptive Statistics

Once generated, there are two ways that data are analyzed in clinical trials via statistical analysis. The first, descriptive statistics, present, arrange, and summarize data of a variable. Further classified into measures of central tendency and variability, these groupings show information on the appearance and distribution of a set of data. Measures of central tendency, most commonly mean, median, and mode, are used to numerically measure the distribution of data. When selecting the appropriate measure of central tendency, one must consider the type of data (Table 10.6).

Measures of variability are used to denote the spread of a set of data. Standard deviation, range, and interquartile range are common measures of variability. Again, the type of data dictates the selection of the appropriate measure (Table 10.6).

Inferential Statistics

The other method for analyzing data is through inferential statistics. Used to make interpretations about a set of data, specifically to determine the likelihood that a conclusion about a sample is true, inferential statistics identify differences between two groups or an association of two groups; the former is more common in the pharmaceutical literature. Inferential statistics requires the performance of statistical tests to see if a conclusion is correct compared with the probability that conclusion is due to chance. These tests calculate a *P*-value that is then compared with the probability that the results are due to chance. This is alpha (α), which is most often 0.05; therefore, a *P*-value less than 0.05 is typically considered statistically significant.

Power

The other factor of necessity when using inferential statistics is power, the ability to detect a difference if one truly exists. Power is a key aspect of calculating a sample size for a study. Numerically, power is equal to 1minus beta (β), which is typically between 0.1 and 0.2, meaning power is usually between 80% and 90%. Beyond being influenced by β, power is affected by α, sample size, the magnitude of effect (i.e., hypothesized difference to be detected), and the variance of the outcome variable. Investigators must consider power and sample size to enroll enough participants to detect a difference without wasting resources by over enrolling or treating patients with a potentially ineffective therapy.

Error

Both α and β share a relationship with the potential for error when making a conclusion based on inferential statistics. Type I error is directly related to α and defined as concluding there is a difference when one does not actually exist. The probability of making a type I error is equal to α used in the study; the only reason a type I error could occur is a chance, though because *P*-values are commonly 0.05 or 5%, 1 of 20 times investigators conclude a difference they may be making a type I error. Therefore, the more times a single set of data is analyzed, the higher likelihood a type I error will occur. Type II error is related to β, which occurs when investigators conclude no difference when one actually exists. The probability of making a type II error is equal to the β used in the study; type II errors occur due to chance or because the required sample size was not enrolled. From a readers' perspective, one must ensure that authors report a power calculation so it can be determined if they make a type II error. Looking for α, β, magnitude of effect, and calculated sample size in the methods of the paper is important.

TABLE 10.7 Summary of Univariate Analysis Test Selection

	Independent Groups		Paired Groups	
Data Type	**2**	**3 or More**	**2**	**3 or More**
Nominal	Chi-square or Fisher's exact	Chi-square or Fisher's exact	McNemar test	Cochran Q
Ordinal	Mann-Whitney U or Wilcoxon rank sum test	Kruskal-Wallis	Wilcoxon signed rank test	Friedman test
Continuous, normally distributed	Student's t-test	ANOVA	Paired t-test	Repeated measures ANOVA
Continuous, not normally distributed	Mann-Whitney U or Wilcoxon rank sum test	Kruskal-Wallis	Wilcoxon signed rank test	Friedman test

Univariate Analysis

Inferential statistical analysis can be broken into two broad categories: univariate analysis and multivariate analysis. The univariate analysis uses one dependent variable, the outcome, and one independent variable, the intervention. The selection of the appropriate statistical test is determined based on the answers to a few simple questions. First, the reader must answer two questions about the dependent variable: (1) "Are the data nominal, ordinal, or continuous?" and (2) "Are the data paired or independent?". The reader should then ask "How many groups are being compared?" with the only answer options being either two or more than two. Finally, the reader should determine the intention of the statistical test. The two options are either to detect a difference between groups or describe the relationship between two variables. Statistical tests, correlation, and regression, describing relationships between two groups, are beyond the scope of this chapter. Table 10.7 summarizes the statistical tests used to compare groups based on the type of data of the dependent variable (i.e., outcome). For time-to-event data, Kaplan-Meier survival analysis is used to measure the actual length of time to an outcome, generating survival curves that show how long it took for patients to experience the event. These curves are visual only and require the use of a log-rank test to determine if a statistically significant difference exists between them.

Multivariate Analysis

Another way to use statistical analysis is to control for potential confounding factors. Investigators use multivariate analysis, which determines whether two groups or conditions are different while controlling for confounding variables. This occurs in observational studies but may also be required in randomized controlled trials, most commonly to compensate for any potential influence of study sites in multicenter trials. Similar to univariate analysis, the type of statistical test is chosen based on the dependent variable (Table 10.8).

To summarize, important skills from this section include the ability to identify types of variables and data, a grasp of descriptive statistics, the knowledge of power calculation elements, and an understanding of questions that need to be answered to pick a statistical test. Often, the type of data is not explicitly stated in an article, but once determined, the researcher or the reader can easily pick the correct statistical test. Incorporating these concepts into the knowledge base of a pharmacist will provide an important tool that can be used when evaluating medical literature.

TABLE 10.8 Multivariate Analysis Statistical Test Based on Data Type of Dependent Variable

Type of Data of Dependent Variable	Multivariate Analysis
Nominal or ordinal	Logistic regression
Continuous	ANCOVA
Time-to-event	Cox hazards model

Results

In general, the first thing that will appear in the results section of an article is the baseline characteristics of the sample. This may be accompanied by information on enrollment via a flowchart that includes the subjects enrolled, randomized, receiving medication, completing the study, and analyzed. This flowchart should include reasons participants withdrew from the study, which allows the reader to draw conclusions about the intervention after considering attrition.

Baseline characteristics are information collected on participants prior to randomization and can include demographics, disease states, medications, social history, and other variables that impact the study outcome. Baseline characteristics are used to determine if randomization worked by showing differences between groups; the reader should consider whether the groups are equal at baseline. Baseline characteristics are also used to determine the applicability of the sample to the patient population encountered by the reader. Asking whether the sample represents the patients seen by the reader is good practice.

The focus of reviewing results should be on the statistical significance of the outcomes. If the *P*-value is less than α, then a statistically significant difference is detected. Additionally, confidence intervals (CIs), a range of values that include the upper and lower range of possible values for a population based on a sample, are considered. The CI presents an estimate of observed results that will be seen in the population. The CI is typically presented as 95%, which means that with 95% confidence, the true population value is within the CI. As sample size increases, the CI narrows because the investigator is pulling more observations out of the population into the sample. This is largely under the investigator's control, whereas the other major CI influencer, the spread of data, is not. As data become more spread out around a measure of central tendency, the CI will widen.

The CI has two main uses. It is preferable to the *P*-value in presentation of results because it provides information on the effect size in addition to statistical significance. To determine statistical significance from a CI, first consider the type of comparison being made. If a ratio is being calculated, rates of an outcome are divided by one another; if the CI includes 1, then the groups are equal and the result is not statistically significant. Common ratios seen in the medical literature include odds ratios, hazard ratios, and relative risk. For non-ratio data, when groups are being compared via subtraction, the point of no difference is 0; a common example is a comparison of the mean difference between two interventions, which are continuous data (e.g., two antihypertensive agents being evaluated for blood pressure reduction).

Baseline data, enrollment information, and statistical significance will help the reader conduct a power analysis, which is a method to determine if the study has "met power," which means the investigators enrolled enough patients so that if they see no difference, that conclusion is valid. Answering questions via a power algorithm (Fig. 10.2) is a simple way to conduct a power analysis. If the study concludes a significant difference, then power ceases to be a consideration. If no difference is seen, then the reader should determine whether all of the elements of a power calculation were reported. Additionally, reviewing that the conclusion being drawn is related to the outcome used to identify the sample size, usually

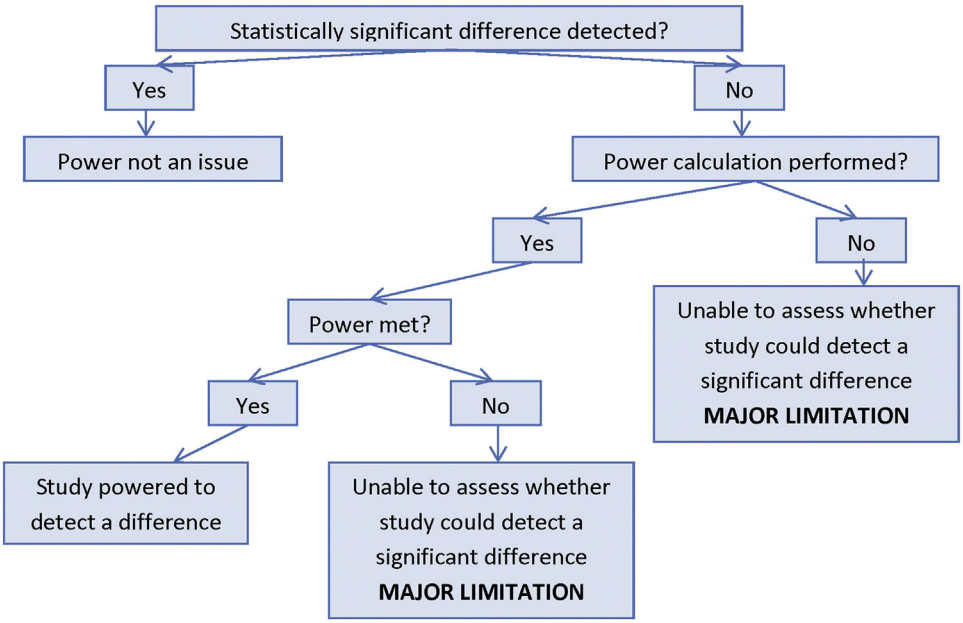


FIGURE 10.2 Power analysis algorithm to appraise power of a study population.

TABLE 10.9 Contingency Table

	Event	No Event
Intervention	A	B
Control	C	D

the primary outcome, is important. This means that results for secondary and safety outcomes are likely to be underpowered; conclusions of no difference for these outcomes are prone to a type II error. Assuming the conclusion is related to the outcome on which the power calculation was based, the reader should check to see if the number of patients or events required to detect a difference were enrolled or occurred, respectively. If the answer is yes, then the study is powered to detect a difference, and the conclusion of no difference is valid. If not, then there may be a type II error, and the conclusion of no difference is invalid.

Once statistical significance is evaluated, and the conclusion is determined appropriate, a reader should consider clinical significance. For continuous and ordinal data, one must consider the range of plausible results for the outcome, the absolute difference, and whether the intervention helps patients meet treatment goals.

For nominal data, measures of association are calculated. Often, authors report the most beneficial value to support their case. Readers need to interpret that value and calculate other values that better represent the difference observed. The first step to calculating measures of association is to develop a contingency table (Table 10.9). One can use a simple example of 20 patients randomized to receive aspirin (n = 10) or placebo (n = 10) to prevent MI over the next 10 years to calculate measures of association. One patient and three patients have MIs in the aspirin and placebo groups, respectively. This is equal to a 10% (1/10) risk in the aspirin group and 30% (3/10) risk in the placebo group.

From risk, one can calculate the four measures of association: relative risk (RR), relative risk reduction (RRR), absolute risk reduction (ARR), and number needed to treat (NNT). RR, sometimes called risk ratio, is the risk associated with exposure on disease state development. It is calculated by dividing the exposed group risk by the unexposed group risk. In this case, 10% divided by 30% is equal to 33%; this means the risk of an MI in the aspirin group is one-third the risk of an MI in the placebo group. RRR is the percent of risk removed by introducing the intervention and is calculated by subtracting RR from 100% because everyone has an equal baseline risk. In this example, RRR is 100% minus 33%, which equals 67%; a person's risk of an MI is reduced 67% by taking aspirin instead of placebo. ARR represents the difference in the percentage of subjects developing an outcome in the control group in comparison with the intervention group. This is calculated by subtracting the risk of the outcome in an unexposed group by the risk in an exposed group. Subtracting the 10% risk in the aspirin group from the 30% risk in the placebo group yields a value of 20%, meaning that 20% of participants were spared an MI due to aspirin. NNT represents the number of patients that need to receive an intervention to prevent one event. Calculated as the reciprocal of ARR, one would divide 1 by 20%, or 0.2, to get a value of 5; therefore, to prevent one MI, five patients need to receive aspirin.

The most valuable measure of association in the determination of clinical significance is dependent on the outcomes and results; however, clinical studies will often present the most appealing value, usually RRR. Readers need to calculate for themselves the measures of association not presented by the authors. A 67% RRR may appear more impressive than a 20% ARR, but the latter is likely more indicative of the clinical value of the intervention. With small sample sizes, these numbers are closer together, but as sample size increases, the relative and absolute differences will become much more disparate.

Beyond efficacy data, safety results must also be considered. All articles should report adverse event data. Because adverse event incidence is often small, readers need to make sure not to draw sweeping conclusions from the results. Adverse events are an excellent example of an opportunity for a type II error because the study would not have been powered to detect a difference in the rates of these outcomes, yet authors might suggest that there was no greater incidence of adverse events for the intervention. This concludes no difference exists when one might because the study was not powered to detect that difference.

Safety results are also important to consider when determining clinical significance. A major consideration of therapeutic decision-making is that the benefits of an intervention outweigh the harm. Considering the NNT and comparing it with a similar calculation specific to adverse effects, the number needed to harm (NNH) is helpful. The NNH is calculated by determining the absolute risk increase (ARI) of an intervention. If two patients taking aspirin developed a gastrointestinal bleed (GIB), whereas only one patient taking placebo had a similar event, the risk of GIB is 20% and 10%

in the aspirin and placebo groups, respectively. ARI is calculated by subtracting the rate of GIB in the control group from the intervention group, in this case, 20% minus 10% equals 10%, meaning that taking aspirin increases the risk of having a GIB by 10%. Similar to NNT, NNH is calculated by taking the reciprocal of the ARI, so 1 divided by 10%, or 0.1, equals 10; 10 patients would need to receive aspirin to cause one GIB.

While NNT and NNH can be valuable for determining the benefit−risk ratio of an intervention and thus clinical significance, it is important to understand that these results have no meaning if a statistically significant difference is not identified and there is no way to define a "good" NNT or a "bad" NNH. Readers should review the difference between the numbers, the severity of the event (e.g., nausea vs. death) and the length of the trial.

Readers may also be required to evaluate the results of systematic reviews or metaanalyses, as the use of these study designs is increasing.[54,55] For metaanalyses, results are often presented using a forest plot, which graphically displays the results of included studies along with a combined effect size.[55] An example forest plot is shown in Fig. 10.3. Each of the six studies in this example has an effect estimate, represented by a block, and corresponding CI, denoted by the horizontal line through the block. The size of the block represents the influence that each study has on the overall results, which is also referred to as study weight. Studies with more precise estimates are given more weight and represented by larger blocks in the forest plot. The diamond is the effect size and corresponding CI when the results of all six studies are combined. One can determine if the results are statistically significant by evaluating if the CI crosses the line of no effect, which is 1 and 0 for the ratio (as shown in Fig. 10.3) and non-ratio data, respectively. By inspecting the forest plot, the reader can quickly determine to what degree the results of individual studies differ and identify studies with the greatest impact on the overall results.

Important steps for readers to take when considering results include looking at the baseline characteristics to make sure randomization worked and that the sample represents the patients seen by the reader. Although performing a power analysis and interpreting statistical significance via *P*-values or CIs are also an important skill, the most important skill is to take statistically significant results and determine clinical significance. Developing this skill set will take time because there is a great deal of subjectivity associated with clinical significance, but appropriately interpreting results and being able to calculate measures of association will reduce this subjectivity.

Discussion

The discussion section is an opportunity for the authors to provide their interpretation and assessment of the results. This section should summarize key findings, compare and contrast results with previous studies, present strengths and limitations, mention future investigations, and end with a conclusion. As a reader, making sure that authors do not exaggerate the results is important. Common flaws of discussion sections include authors only citing studies with similar

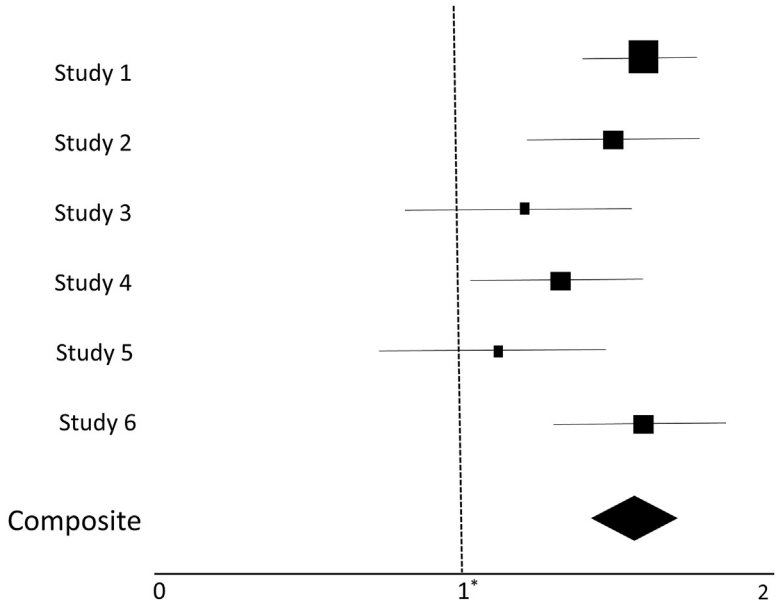


FIGURE 10.3 An example of forest plot.

results, failing to discuss limitations, and making inappropriate conclusions not related to study results. Readers should take caution when reviewing this section, never relying on it to make therapeutic decisions.

Conclusion

The conclusion of the article will be a separate section after the discussion or appear as the last paragraph of the discussion. This is the authors' final determination and clinical recommendation of the evidence presented. The conclusion should focus on primary outcome results and be tied directly to the primary objective of the study. It should not address information not presented within the article itself. As a reader, ensure that the paper and abstract conclusions match, if not in words, at least in intention. To summarize the overall theme of this chapter, one must make their own decisions from an evaluation of the full paper and not just from any one section the authors present, including the conclusion.

Bibliography

The bibliography of the article will follow the conclusion; authors document references used to support the information contained within the article. Any work included in the bibliography should be internally cited within the manuscript. A good reader of the medical literature will scan the bibliography to make sure authors used materials that were from reputable sources and that they did not extensively cite their own work. The bibliography is also a great tool from which to pull cited literature to verify that the documented information is accurate and to gather additional information on the topic.

OBJECTIVE 10.3. UTILIZE CLINICAL EXPERTISE TO IDENTIFY THE OVERALL VALIDITY OF A GIVEN CLINICAL TRIAL

After reviewing every section of an article, one should consider the overall validity. The first consideration is to determine the statistical significance of the results.[8,56–59] If the results are significant, validity should then be considered; if not, the study may prove valuable to review for a future hypothesis generation, but clinical decisions may not be necessary.

Internal validity is the degree to which a study establishes a causal relationship between the treatment and an observed outcome, whereas external validity is the degree to which the results of a study can be generalized to patients outside of the study sample. Both are important considerations when reviewing the medical literature and can be compromised in several ways.[58,60]

Threats to internal validity are detailed in Table 10.10. Most threats, except for experimental mortality, can be removed with randomization. Randomization ensures that threats to internal validity will be present in all study groups; thus, if the results are statistically significant, differences occurred in spite of threats to internal validity. Because of attrition after randomization, ITT analysis is the best way to control for experimental mortality. To assess internal validity, it is important to analyze the method of randomization, appropriateness of blinding, the rate and reasons for experimental mortality, and if scientific misconduct occurred.

Threats to external validity include the Hawthorne effect, which occurs when participants behave differently knowing that they are being observed.[59,60] Examining baseline characteristics of a sample for representativeness of the population is another consideration. Overly strict inclusion and exclusion criteria or inappropriate sampling can cause results of a study to have limited external validity.[56,58–60] Finally, sometimes studies involve complex treatment regimens that cannot be controlled outside of the study environment; in these cases, results seen in the study may not be achievable in the population.[59,60] To analyze external validity, it is important to assess the appropriateness of sampling methods, the inclusion and exclusion criteria of a study, baseline characteristics, and adherence to the study protocol.[56–59]

Assessing the validity of a study helps the reader determine whether the results of a single study are true and can be applied outside of the study sample. There are many threats to internal and external validity, so it is important to take a standardized approach to assessing the possible presence of these threats. Ultimately, assessing validity does not lead to an absolute decision about whether to apply evidence to practice but allows the reader to analyze the degree to which one should consider using the evidence (e.g., will the ultimate recommendation apply to all patients or just patients similar to those studied?).

TABLE 10.10 Threats to Internal Validity[58–60]

Threat	Definition	Example
History	External time-related factors affect the dependent variable and bias results.	Legislation requiring the use of seat belts could confound an experiment designed to determine whether an intervention reduces motor vehicle mortality.
Maturation	Study subjects change over time due to natural progression.	Childhood asthma may resolve on its own or as a result of a specific study intervention.
Testing	Testing subjects multiple times with the same assessment tool may capture improvement not related to the intervention.	Pre- and postassessments of learning that contain the same questions may show improvement simply because the participants were aware of the questions and paid closer attention to those topics.
Instrumentation	Outcomes assessed by measurement tools or devices may change in response to changes in the tool and not related to the intervention.	A new blood pressure cuff is purchased that measures blood pressure more accurately. Study subjects' blood pressure may appear to have improved simply because the tool changed and not as a result of the intervention.
Statistical regression	Subjects selected based on extreme values may show improvement simply because extreme values tend to regress toward the mean and not because the treatment is effective.	A clinical trial for a blood pressure medication only includes patients with excessively high blood pressure. Some patients' blood pressure may improve simply because the value cannot increase any further whether or not the patient receives the medication.
Selection	How subjects are divided into study groups creates groups that are unequal with regard to variables that can affect the study outcome.	A cohort study is performed in two clinics. One is a military-based clinic and the other serves primarily elderly patients. If each clinic serves as a cohort, the groups will not be equal. One group may be mostly young men (i.e., military-based clinic) and is compared with mostly elderly patients.
Experimental mortality	Patients lost throughout the study period (i.e., attrition, withdrawal, dropout) are not lost equally between groups.	A study drug has a terrible side effect, so more patients receiving the treatment drop out of the study. The final group of patients that are analyzed to show the drug is very efficacious; however, that effect will only be realized in patients that can tolerate the terrible side effect and continue taking the medication.

CONCLUSION

As one can see, there is a great deal to consider when reading an article. All studies have some bias, and it is the job of the clinician to determine to what degree that bias will impact their ability to use the information in the study. Over time, considerations described in this chapter will become natural for consumers of the literature. The more reading and interpretation one does, the more flaws they see and the more questions they ask. Using this chapter as a guide for things to consider in MLE will help future and current clinicians as they continue to develop these skills and thus their ability to help their patients.